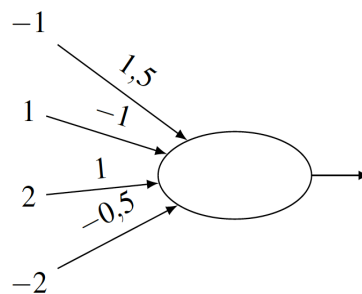




Exercise 1

Activation Functions - SOLUTION

Calculate the output of the following neuron for each of the following activation functions:



Let's first calculate the weighed sum x : $x = (-1)*1.5 + 1*(-1) + 2*1 + (-2)*(-0.5) = 0.5$

a) Step function.

```
if(x>0):  
    print("1")  
else:  
    print("0")  
→ 1
```

b) Sigmoid.

```
print(1/(1+math.exp(-x)))  
→ 0.6224593312018546
```

c) ReLU.

```
if(x<=0):  
    print(0)  
else:  
    print(x)  
→ 0.5
```